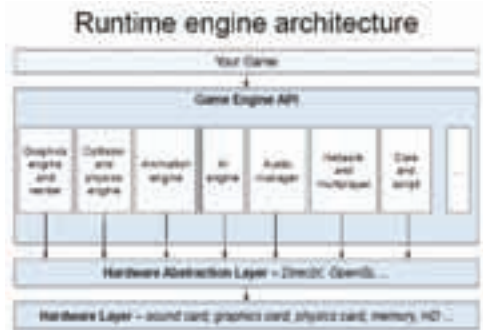


- ▶ The Graphic Engine takes care of low-level optimized scene graph exploration and rendering, visual effects and front end HUD, menus, GUI.
- ▶ Handle shadowing and Lighting in the Game world.

Collision and Physics engine

- ▶ Handles simulations of the game world like physical Behavior, Collisions, Terrain changes, Explosions, Object breaking and destruction.
- ▶ Physics Engine takes care of physics-based animations.
- ▶ Some famous physics Engine SDKs include – Havok, ODE (Open Dynamic Engine), Tokamak, PhysX.



Game engine architecture

Animation engine

- ▶ It is partially linked to physics Engine.
- ▶ It handles off-line motion capturing and retargeting, motion editing and annotation.
- ▶ It handles real-time sprite/texture animation, vertex animation, rigid body and skeletal motion.
- ▶ Some of the famous Animation Engines are: Granny (majorly used in games), Havok Animation, Edge (for PS3)

AI engine

- ▶ Handles the behavior and Interaction of all the NPCs in the game.
- ▶ Handles all the spatial displacement like obstacle avoidance, path planning.
- ▶ Creates the base code architecture for scripting Strategies in the game world like Hiding, Attack/Defense.
- ▶ Handles the Decision making of NPCs in the game.
- ▶ Some famous AI game Engines include AI Implant by Presagis, Kynapse by Autodesk, DirectIA by MASA, SimBionic.